

SOORIYA ARACHCHIGE CHIRANTHA AKARSHA

MECHANICAL ENGINEERING UNDERGRADUATE (MECHATRONICS SYSTEMS ENGINEERING STREAM)

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PROFILE / SUMMARY

3rd-year undergraduate studying Mechanical Engineering with a specialization in Mechatronics Systems Engineering at the University of Moratuwa. Passionate about industrial automation, autonomous systems design, robotics, embedded systems, and control systems engineering. Looking to secure a mechatronics engineering internship to apply analytical skills and hands-on experience to solve real-world engineering challenges.

EDUCATION

BSc Mechanical Engineering (Mechatronics Systems Engineering Stream) 2023 – Present
University of Moratuwa

- **Relevant coursework:** Object Oriented Programming for Mechanical Engineers (A+), Mechatronic Systems Engineering (A), Sensors / Actuators and Intelligent Systems (A), Instrumentation System Design (A-).
- **Current Coursework:** Robotics, Embedded Systems, Control Systems, Mechatronic System Design Project.

Ananda College, Colombo

Advanced Level Results 2022

- **Results:** Combined Mathematics (A), Physics (A), Chemistry (A), General English (A).
- **Rankings:** Island Rank: 426, District Rank: 108.

Thurstan College, Colombo

Ordinary Level Results 2019

- **Results:** 9 A's (Distinction passes) including Information & Communication Technology (ICT) and Business & Accounting Studies.

Other Certifications

- NVQ Level 4 in Software Development — Vocational Training Authority, NVTI, Narahenpita (12-month full-time course).

SKILLS

- **Technical / Field-Specific:** Hardware design, simulation, analysis, and debugging; circuit and electronic systems design and debugging; basic manufacturing and rapid prototyping.
- **Software:** SolidWorks, AutoCAD, Altium, Proteus, KiCAD, MATLAB, ADAMS, RapidHarness, Ansys, ROS.
- **Programming:** C++, Python, Java.
- **Languages:** Sinhala (Native), English (Fluent).
- **Transferable:** Problem-solving, fast learning, innovative thinking.

PROJECTS

FalconE Racing — Formula Student Electric Vehicle

Altium, RapidHarness, Soldering, PCB Debugging tools

- **The Problem:** Development of the electronics subsystem for an electric formula race car competing internationally.
- **Contribution:** Designed and developed the vehicle wiring harness and custom PCBs. Built custom testing apparatuses for IMD and BSPD. Managed the 600V high-voltage battery system and debugging process to ensure strict competition rule compliance.
- **Tools:** Altium, KiCAD, RapidHarness, Soldering(SMD, TH), PCB Debugging tools.
- **Outcome:** Successfully competed as a team at Formula Student Spain 2026.

Rho Solutions (Startup) — Automated Rapid Prototyping Platform

Python (Machine Learning), Web Development, 3D Printing Software

- **The Problem:** Co-founded a startup requiring a streamlined digital infrastructure for instant customer quoting and order processing.
- **Contribution:** Built a full-stack web application (rhosolutions.lk) for order tracking and invoicing. Engineered a machine learning model (utilizing Random Forest/XGBoost trained on the Thingi10K dataset) to instantly estimate 3D printing prices directly from STL geometry without requiring slicing software.
- **Tools:** Python (Machine Learning), Web Development, 3D Printing Software.
- **Outcome:** Launched a fully functional business platform automating the customer workflow.

Project Diyakawa — Autonomous Underwater Vehicle (SAUVC 2026)

SolidWorks, Firebase, C++, Microcontrollers

- **The Problem:** Designing an AUV for the Singapore AUV Challenge in Sanya, China.
- **Contribution:** Designed the electronic architecture and motor brackets. Developed a Firebase-based parameter tuning system with custom microcontroller firmware to control the vehicle's navigation.
- **Tools:** SolidWorks, Firebase, C++, Microcontrollers.
- **Outcome:** Successfully passed the qualifying round and secured a spot in the on-site competitions.

Tree-Climbing Robot — Mechatronics Systems Design (Ongoing)

CAD, Kinematic Simulation

- **The Problem:** Developing a robot capable of reliably ascending tapered and inclined coconut palm trunks without damaging the tree.
- **Contribution:** Designing and simulating an autonomous climber utilizing active taper adaptation, tensioned band grippers, and Time-of-Flight sensors for inherently stable movement.
- **Tools:** CAD, Kinematic Simulation.
- **Outcome:** Successfully advancing through the design and simulation phases.

SCARA Object-Picking Robot — Vision-Guided Sorting System

Arduino, Raspberry Pi, Image Processing, Web Interface

- **The Problem:** Automating the identification and sorting of objects based on shape and color via a user interface.
- **Contribution:** Engineered the mechanical and electronic systems. Programmed an Arduino controller linked to a Raspberry Pi for image processing, implemented robotic kinematic calculations, and integrated an electromagnet end-effector.
- **Tools:** Arduino, Raspberry Pi, Image Processing, Web Interface.
- **Outcome:** Delivered a functional prototype controlled via a custom web interface.

Bicycle Instrumentation and Cloud Logging System

Microcontrollers, ESP-NOW, Firebase, Web Development, Sensors

- **The Problem:** Designing a comprehensive system to measure, process, analyze, and display bicycle telemetry including distance, speed, acceleration, orientation, and power input.
- **Contribution:** Managed orientation and acceleration sensors, implemented ESP-NOW protocol for communication across three microcontrollers, and developed Firebase data logging with a web-based display dashboard.
- **Tools:** Microcontrollers, ESP-NOW, Firebase, Web Development, Sensors.
- **Outcome:** Successfully integrated a three-tier microcontroller system with live cloud syncing and a local display module.

Assistive Safety System for the Visually Impaired

SolidWorks

- **The Problem:** Developing a reliable spatial awareness system to assist blind individuals in navigating their environment.
- **Contribution:** Designed a creative wearable enclosure that has rough shape of a VR box which includes Raspberry Pi with a Camera module, Ultrasonic sensor, Speaker and Its power system.
- **Tools:** CAD(SolidWorks)
- **Outcome:** Ergonomic designs for assistive devices.

Drum Throne Reverse Engineering and Redesign

Mechanical Analysis, CAD

- **The Problem:** Improving the structural integrity and stability of an existing drum throne manufacturing design.
- **Contribution:** Analyzed the existing two-leg assembly and engineered a proposed double-braced leg assembly to enhance mechanical support.
- **Tools:** Mechanical Analysis (ANSYS), CAD.
- **Outcome:** Delivered a finalized double-braced manufacturing design proposal.

Automatic Check-Weighing and Analytics Conveyor System (Ongoing)

Operations Management, Systems Engineering

- **The Problem:** Designing an in-line conveyor-based check-weighing and production analytics system for packaged flour bags.
- **Contribution:** Evaluating and selecting appropriate weighing machine mechanisms to integrate into the automated production line.
- **Tools:** Operations Management, Systems Engineering.
- **Outcome:** Advancing through the component selection and design phase.

Electronic Enclosure Design

3D Modeling, CAD

- **The Problem:** Creating robust protective housings for custom electronic assemblies.
- **Contribution:** Designed functional electronic enclosures tailored to specific PCB dimensions and user interface requirements.
- **Tools:** 3D Modeling, CAD.
- **Outcome:** Produced custom hardware enclosures ready for rapid prototyping.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Team Member, Electronics Subsystem

FalconE Racing (Formula Student Team)

- Participated actively in the design, development, and debugging of competitive electronics systems for the Formula Student international racing team.

Co-Founder

Rho Solutions (Startup)

- Launched and co-managed an entrepreneurial rapid prototyping and 3D modeling solutions business alongside academic studies.

ACHIEVEMENTS

- **Dean's List (Semester 2)** — Faculty of Engineering, University of Moratuwa.
- **Overall 28th Place** — Formula Student Spain (FSS 2026), Circuit de Barcelona-Catalunya (FalconE Racing).
- **On-Site Competition Qualifier** — Singapore Autonomous Underwater Vehicle Challenge (SAUVC 2026), Sanya, China (Project Diyakawa).
- **All-Island Rank 426 (District Rank 108)** — G.C.E. Advanced Level Examination 2022 (Physical Science Stream).

REFEREES

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